

Q8		
(a)(i)	From graph, when $v = 25 \text{ m s}^{-1}$, $P = 500 \text{ W}$	B1
(a)(ii)	$P = Fv$ $F = \frac{P}{v} = \frac{500}{25} = 20 \text{ N}$	M1 A1
(a)(iii)	At constant velocity, the net force on the rider is zero. Furthermore, $F_{\text{slope}} = 0$ since the ground is level. Hence, the propulsive force = the drag force F_{air} $\rho = 1.0 \times 10^{-3} \text{ g cm}^{-3} = \frac{1.0 \times 10^{-3} \times 10^{-3} \text{ kg}}{10^{-6} \text{ m}^3} = 1.0 \text{ kg m}^{-3}$ $F_{\text{air}} = \frac{1}{2} \rho C_D A v^2$ $20 = \frac{1}{2} (1.0) (C_D A) (25^2)$ Effective drag area, $C_D A = 0.064 \text{ m}^2$ C1 – appreciate that $F_{\text{air}} = F$ allow mark as long as 20 N is substituted for M1 – for correct conversion of density to kg m^{-3} A1 – correct calculation of drag area.	C1 M1 A1
(b)(i)	$\tan(\alpha) = \frac{37}{100} \Rightarrow \alpha = 20.304^\circ$	M1
	$F_{\text{slope}} = mg \sin \alpha$ $= (85)(9.81)(\sin 20.304^\circ)$ $= 290 \text{ N}$	M1 A1
(b)(ii)1.	work done against gravity = $(mg \sin \alpha)(x) = (289)(6.4) = 1850 \text{ J}$	M1 A1
(b)(ii)2.	Since the cyclist and bicycle is moving up the slope at constant speed, $F'_{\text{prop}} = F_{\text{air}} + F_{\text{slope}}$	C1
	From (a), $F_{\text{air}} = 20 \text{ N}$	
	Hence, $P' = F'_{\text{prop}} v = (20 \text{ N} + 289 \text{ N}) (25 \text{ m s}^{-1}) = 7725 \text{ W} = 7700 \text{ W}$	M1 A1
(c)(i)	By Newton's 2nd Law, $\Sigma \vec{F} = m\vec{a}$ ($\uparrow$) $N_1 + N_2 = W = mg \dots\dots(1)$ ($\rightarrow$) $\mu N_1 + \mu N_2 = ma \dots\dots(2)$ Hence, $\mu(N_1 + N_2) = \mu mg = ma$	M1 M1

	Solving, $a = \mu g = (0.37)(9.81) = 3.63 \text{ m s}^{-2}$	A1
(c)(ii)	Taking moments about the CG, by principle of moments Sum of anticlockwise moments = Sum of clockwise moments $(43)N_2 + (114)\mu N_1 + (114)\mu N_2 = (107 - 43)N_1$ $\Rightarrow 43N_2 + 42.18W = 64N_1 \quad \dots\dots(1)$ Since, $N_1 + N_2 = W \quad \dots\dots(2)$ Substituting (2) into (1) and solving, $43(W - N_1) + 42.18W = 64N_1 \Rightarrow N_1 = 0.7960W = 0.80W \text{ (Shown)}$	M1 M1
(c)(iii)	Comparing the normal contact force at the front wheel to that at the back wheel, the ratio = 4.	B1
(c)(iv)	If the centre of mass moves forward, the <u>(clockwise) torque produced by N_1 decreases and (anticlockwise) N_2 increases.</u> Hence, there is now <u>a net anticlockwise torque on the bicycle</u> causing the bicycle to flip forward.	B1 B1